Microsoft Elevate

POWER BI PROJECT

AIR QUALITY DATA ANALYSIS AND VISUALIZATION

PRESENTED BY : KESHAV SHARMA

STUDENT NAME: KESHAV SHARMA

COLLEGE NAME: SHRI RAM GROUP OF COLLEGES

DEPARTMENT: BTECH- ELECTRICAL ENGINEERING

EMAIL ID: KESHAV5436101@GMAIL.COM

OUTLINE:

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- System Development Approach
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PROBLEM STATEMENT:

- Air pollution is one of the major environmental challenges affecting public health and urban sustainability. Cities generate large volumes of air quality data, but this data is often difficult to interpret and analyse without proper visualization tools.

- Traditional reports and raw datasets do not provide quick insights into pollution trends, peak pollution hours, or seasonal variations.

- Therefore, there is a need for an interactive data visualization dashboard that can analyse air quality data and present meaningful insights for decision-makers and the public.

PROPOSED SOLUTION:

The proposed solution is to develop an interactive Air Quality Dashboard using Excel, Power Query, and Power BI.

The dashboard will:

- Collect and organize air quality data from datasets.
- Clean and transform the data using Power Query.
- Perform data analysis and create calculated metrics.
- Visualize pollution patterns using Power BI dashboards.
- The dashboard will allow users to:
- Monitor pollution trends over time
- Identify peak pollution hours
- Compare pollutant levels
- Analyse monthly and daily air quality patterns

SYSTEM APPROACH:

System Requirements

Hardware:

- Laptop / PC
- Minimum 8GB RAM

Software:

- Excel
- Power BI

Libraries Required

- Not used as it is a simple Project and Not used ML or Python

Tools Used

- Power BI for dashboard visualization
- Excel for dataset storage

ALGORITHM & DEPLOYMENT:

Algorithm Selection:

The process includes:

- **Data Extraction**
- **Data Cleaning**
- **Data Transformation**
- **Data Modelling**
- **Interactive Visualization**.

Data Input : The input dataset contains historical **air quality and environmental parameters**.

Input features used in the analysis include: Timestamp (Date & Time), PM2.5,PM10, CO, NO_2, SO_2, O_3, Temperature, Humidity, Pressure, Air Quality Index (AQI)

These variables help analyse pollution trends and environmental conditions.

Prediction Process : Users can analyse pollution trends by:

- Time
- Pollutant type
- Monthly variation
- Hourly pollution levels
- These insights help identify **peak pollution hours and seasonal pollution patterns**.

RESULT:

The developed Power BI dashboard provides clear insights into air pollution patterns.

Key observations from the analysis:

- Average AQI observed: around 250 (Poor category)
- Highest PM2.5 concentration recorded: ~200 µg/m³
- Peak pollution hours identified during early morning and late evening
- Seasonal variations observed with higher pollution levels during summer months
- The dashboard enables interactive exploration of air quality trends, helping users quickly understand pollution patterns.

CONCLUSION:

The project successfully demonstrates the use of Power BI and Power Query for environmental data analysis.

The dashboard provides:

- Clear visualization of air quality trends
- Interactive exploration of pollution data
- Easy identification of pollution peaks and seasonal patterns

FUTURE SCOPE:

The system can be enhanced by:

- Integrating real-time air quality data APIs
- Expanding the dashboard to include multiple city datasets
- Adding mobile-friendly dashboards
- Incorporating predictive analytics for AQI forecasting

REFERENCES:

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2. Power Query Documentation – Microsoft Learn
   https://learn.microsoft.com/power-query/

3. Microsoft Excel Documentation
   https://support.microsoft.com/excel

4. Air Quality Index – Kaggle

6. Microsoft Learn – Data Analysis with Power BI
   https://learn.microsoft.com/training/powerplatform/power-bi/

GitHub Link:

Thank You